SG No. 186 C_{6v}^4 $P6_3mc$ [hexagonal]

* plus set: $+[0,\ 0,\ 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `3m.`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,7,8,9]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4,5,6,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3,7,8,9]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[4,5,6,10,11,12]`

Table 3: Wyckoff site: `6c`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ -x,\ z]$	`[1,7]`
2	$[x,\ 2x,\ z]$	`[2,9]`
3	$[-2x,\ -x,\ z]$	`[3,8]`
4	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	`[4,10]`
5	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	`[5,12]`
6	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	`[6,11]`

Table 4: Wyckoff site: `12d`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`
3	$[-x+y,\ -x,\ z]$	`[3]`
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`
5	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	`[5]`
6	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	`[6]`
7	$[-y,\ -x,\ z]$	`[7]`
8	$[-x+y,\ y,\ z]$	`[8]`
9	$[x,\ x-y,\ z]$	`[9]`
10	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	`[11]`
12	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	`[12]`